

Relationship between speech and language skills in young children with nonsyndromic cleft palate with or without cleft lip: A meta-analysis

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Author Introduction



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Overview

What is NCP/L?



Speech & Language
Characteristics



Methods



Results



Clinical Implications

Nonsyndromic cleft palate with/without cleft lip

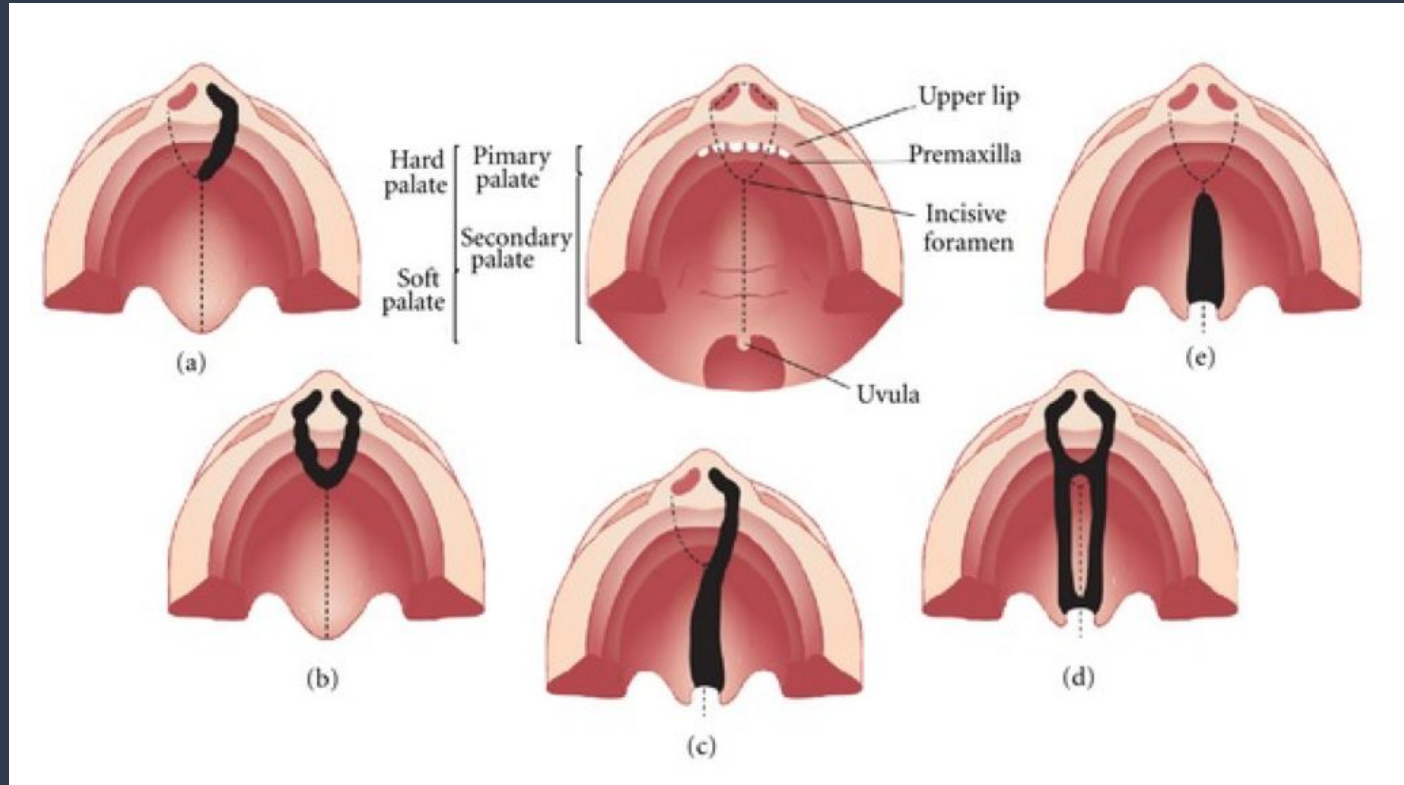


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Speech & Language



**Consonant
Inventory**



**Speech
Accuracy**



**Language
Function**

Language Impact



Purpose

**Describe the average correlation
between early speech and language
skills for young children with cleft
palate**

Methods

A dark blue, diagonal, triangular shape that originates from the bottom-left corner and extends towards the top-right corner, covering the lower half of the slide.

Search

**PubMed + PsycINFO
n = 814**

Hand Search n = 4

Screening

**Title/Abstract
Review n = 818**

Excluded n = 579

**Full Text Review
n = 239**

Excluded n = 208

Scoring

**Studies reporting
speech and language
data n = 31**

**Studies yielding
correlations or data
necessary for
computing
correlations
n=7**

Inclusion Criteria

**Presented data on children
with NSCP/L**

**No diagnosed syndrome or
additional congenital anomaly**

Birth - 8 years of age

Group study design

**Speech articulation measure
+ language measure**

**Available in online journal or
interlibrary loan**

Exclusion Criteria

Not written in English

Single subject study design

**Correlation could not be
retrieved from the original
text article and the
correlation data was not
provided upon request**

**Study
Characteristics**

**Sampling
Characteristics**

**Measurement
Characteristics**

**Quality
Characteristics**

Study Characteristics

Publication Status

Publication Type

Study Design

Study Location

**Analytic + Data Reporting
procedures used**

Sampling Characteristics

Sampling Procedure

Sample Size

**Percentage of male
participants**

VPI function presented

Cognitive Testing performed

Inclusion of comparison group

Type of comparison group

Measurement Characteristics

of speech articulation variables reported

of speech assessment variables reported

Speech Sampling context

Perceptual assessment present

of expressive/receptive language variables reported

of language tests administered

Method of language testing

Quality Characteristics

Speech reliability

Type of speech reliability

Transcriber experience

Average years of transcriber experience

Language reliability

Type of language reliability

Effect Size Coding

Extracted correlations (r) and sample sizes

Transformed to Fisher's z

Obtained variance estimate & 95% confidence interval

Statistical Analyses

Used metafor in R to conduct a random effects meta-analysis

Model provides information about:

- average correlation

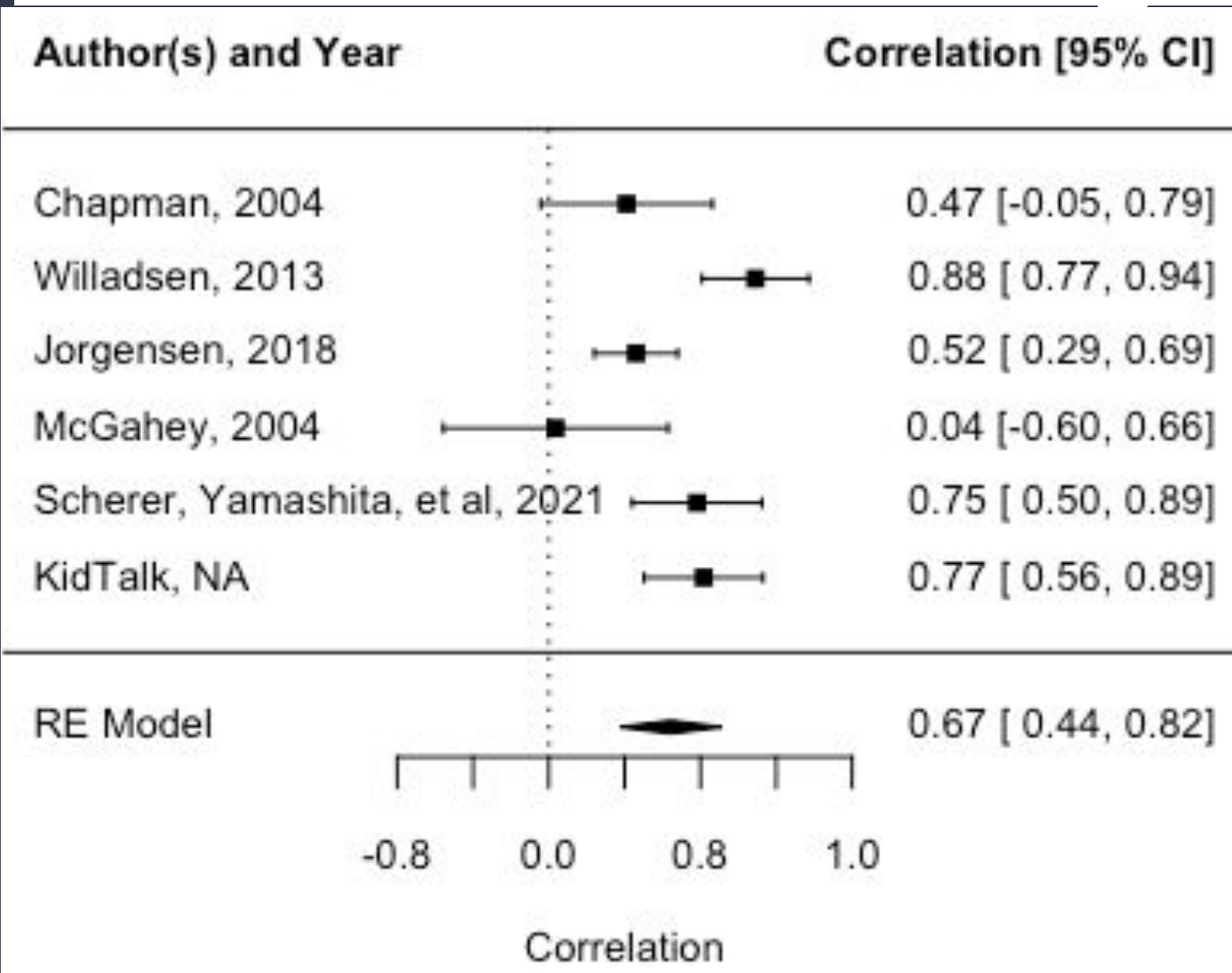
- heterogeneity within & between studies

Back-transform z to r for interpretation

Results

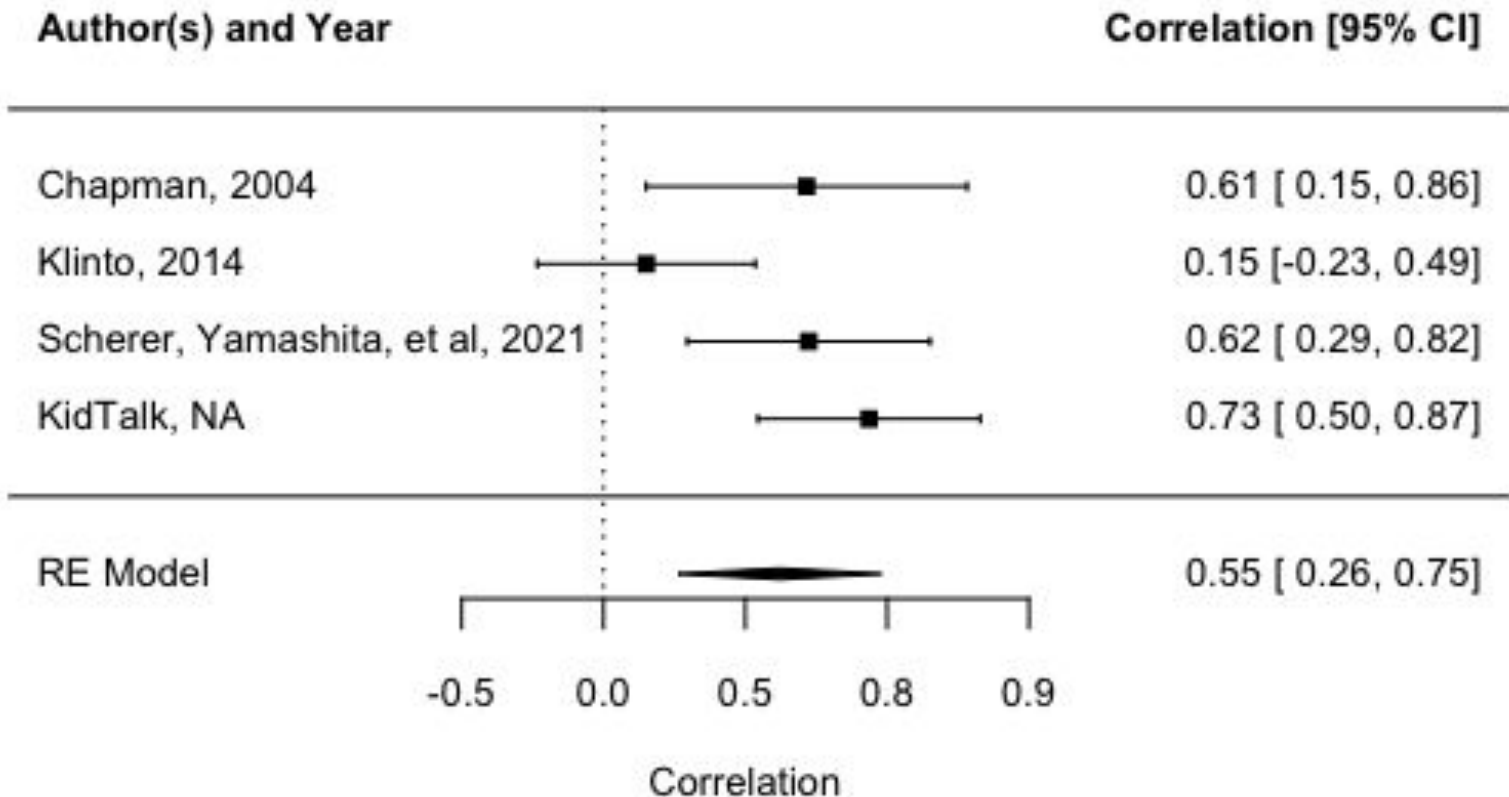
A large, solid dark blue shape that starts from the bottom left corner and extends diagonally upwards towards the right, covering the bottom half of the slide.

Consonant Inventory & Vocabulary



Heterogeneity statistic	Value
Q	18.97 df = 5 p = .002
I ²	74.03 [30.88, 96.53]
τ ²	.125 se = .112

Consonant Inventory & MLU



Heterogeneity statistic	Value
Q	8.547 df = 3 p = .036
I ²	63.12 0, 96.92]
T ²	.083 se = .101

Summary of Results

A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.

**Strong early
speech skills**



**Strong
language skills**

**Speech +
Vocabulary**



**Speech +
Morphology**

Clinical Implications



Have a comment or question?

Feel free to reach out to any of us

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Thank You!

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